

CiviConnect: A SaaS platform Bridging civil sector service groups

Group No: A08

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Abstract

CiviConnect is a SaaS platform designed to enhance civil service delivery by connecting citizens with verified professionals such as plumbers, electricians, and civil contractors. The platform simplifies access to essential services, particularly in developing regions where users often face challenges like unreliable providers, lack of grievance mechanisms, and insecure payment systems. Built using modern technologies like Node.js, Express.js, MongoDB, TailwindCSS, and Python, CiviConnect is a responsive, scalable, and intelligent platform. It supports key features including complaint registration, service booking, real-time chat, secure online payments, and legal agreement generation. To personalize the user experience, the platform integrates machine learning algorithms—such as ranking-based recommendation system focuses on ordering items by relevance to a user, rather than just providing a list of recommendations. CiviConnect ensures efficient, data driven decision-making. The system supports multiple user roles—service seekers, providers, and administrators—ensuring a structured and transparent workflow from request to resolution. Future enhancements include integration with government portals, regional language support, offline accessibility for rural areas, and expansion into civic volunteering and job listing services.

1. Introduction

1.1 Overview

CiviConnect is an online marketplace connecting customers to verified service professionals across domains such as plumbing, electrical work, painting, home cleaning, appliance repair, and more. It modernizes the traditional offline method of finding professionals by providing an intelligent, transparent, and secure digital workflow.

1.2 Problem Statement

Customers face multiple challenges in today's unorganized service sector:

1. Customer posts a request (e.g., plumbing leak).
2. Professionals in the area get notified.
3. Interested professionals send quotations.
4. Customer compares & hires one.
5. Work gets scheduled, tracked, and completed.
6. Payment done via app or cash
7. Customer leaves rating & review.

1.3 Objectives

- Provide a unified platform connecting customers with verified local professionals
- Enable real-time quotation and comparison
- Offer secure, trackable, and scheduled service workflows
- Integrate payment and feedback mechanisms
- Ensure transparency using documentation and review systems

1.4 Scope of the Project

This project covers:

- User and provider registration
- Verified provider onboarding
- Service request creation
- Quotation and negotiation
- Booking and scheduling
- Real-time chat and tracking
- Payment processing
- Rating, review, and dispute resolution
- Admin operations and monitoring

2. System Analysis

2.1 Existing System

The existing service marketplace mostly runs offline or through unorganized local contacts. Customers depend on neighbourhood recommendations or random searches, leading to:

- Unverified professionals
- No accountability
- Inaccurate pricing
- Time delays
- Poor service experience

2.2 Limitations of Existing System

- No structured service categories
- No professional verification
- No mechanism to compare quotations
- No chat, tracking, or scheduling system
- No digital payment or invoicing
- Limited post-service feedback

2.3 Proposed System

CiviConnect solves these limitations by offering:

- Verified professionals with profiles
- Location-based provider discovery
- Transparent quotation mechanism
- Chat and scheduling workflow
- Unified service and transaction records
- Review and rating system

2.4 Key Features

- Customer mobile/web portal
- Provider dashboard
- Admin verification module
- Real-time push notifications
- Payment integration
- Recommendation engine (AI-based)

3. Feasibility Study

3.1 Technical Feasibility

Technologies used:

- **Backend:** Laravel (PHP 8+)
- **Frontend:** HTML5, CSS3, Bootstrap
- **Database:** MySQL

- **Queue:** Redis
- **Deployment:** Docker containers

All components are open-source and widely supported.

3.2 Operational Feasibility

- Simple user interface
- Easy onboarding for service providers
- Scalable for multiple service categories
- Works on mobile and desktop

3.3 Economic Feasibility

Minimal development and hosting cost due to open-source tools.

Long-term revenue streams include:

- Commission per completed service
- Subscription for providers
- Featured listing promotions

4. System Design

4.1 System Architecture

CiviConnect follows a three-tier architecture:

1. **Presentation Layer:** Web/mobile interface
2. **Application Layer:** Laravel REST API
3. **Data Layer:** MySQL database with relational tables

4.2 Data Flow Diagram (DFD)

DFD Level 0:

Customer → Posts request → System notifies providers → Providers quote → Customer selects → Booking → Payment → Review.

DFD Level 1:

- Input module
- Request processing
- Notification service

- Quotation management
- Booking lifecycle
- Payment and invoice module
- Feedback and dispute module

4.3 UML Diagrams (Textual Description)

Use Case Diagram

Actors:

- Customer
- Provider
- Admin

Use Cases:

- Register/Login
- Post request
- Send quotation
- Accept booking
- Make payment
- Submit review
- Verify users (admin)

Sequence Diagram Example:

1. Customer creates request
2. System triggers provider discovery

3. Provider submits quotation
4. Customer accepts quotation
5. Booking created
6. Provider completes service
7. Payment processed

Class Diagram (Key Classes)

- User
- ProviderProfile
- ServiceRequest
- Quotation
- Booking
- Payment
- Review
- Notification

ER Diagram (Textual)

- User(user_id PK, name, phone, role)
- ProviderProfile(profile_id PK, user_id FK, services, rating)
- ServiceRequest(request_id PK, customer_id FK, description, location)
- Quotation(quote_id PK, provider_id FK, request_id FK, price)
- Booking(booking_id PK, provider_id FK, request_id FK, status)
- Payment(payment_id PK, booking_id FK, amount)

5. Module Description

5.1 User & Provider Management Module

Handles registrations, logins, verification, and provider portfolio creation.
Ensures secure identity management.

5.2 Service Request Module

Customer can describe the issue, upload media, and specify location and preferred schedule.

Submit Test report on



Testing: performed on each

model that minimize human intervention & quality of product/decision is enhance.

Till 15/12/2025.

5.3 Provider Discovery & Notification Module

Filters nearby providers based on:

- Service category
- Distance
- Verification status
- Availability

5.4 Quotation Module

human

Providers send quotations with:

- Pricing
- Time estimation
- Material cost breakdown

Customers compare quotations before hiring.

5.5 Booking & Scheduling Module

Manages:

- Booking confirmation
- Rescheduling
- Work progress tracking

AI / DAI

5.6 Real-Time Chat & Tracking Module

Enables customer-provider communication and live location tracking.

5.7 Payment & Invoice Module

Supports:

- Online payments
- Cash recording
- Auto-generated invoices

5.8 Review & Rating Module

Ensures quality control and service accountability.

5.9 Admin Panel Module

Includes provider verification, complaint handling, and dashboard analytics.



5.10 Notification System Module

Sends SMS, email, and push notifications.

5.11 Recommendation Engine Module

Uses ML models to recommend the best provider based on:

- Service category
- Distance
- Rating
- Past success rate
- Response time

5.12 Legal Documentation Module

Creates digital service agreements and receipts.

6. Testing

6.1 Testing Levels

- Unit Testing
- Integration Testing
- User Acceptance Testing
- Security Testing

6.2 Sample Test Cases Table

Test Case	Input	Expected Output	Result
User Registration	Name, Phone	Account Created	Pass
Quotation Submit	Quote data	Quotation Saved	Pass
Booking Confirmation	QuoteID	Booking Created	Pass
Payment	Amount & gateway	Transaction Success	Pass

Confusing matrix
AI
ES

6.3 Security Testing

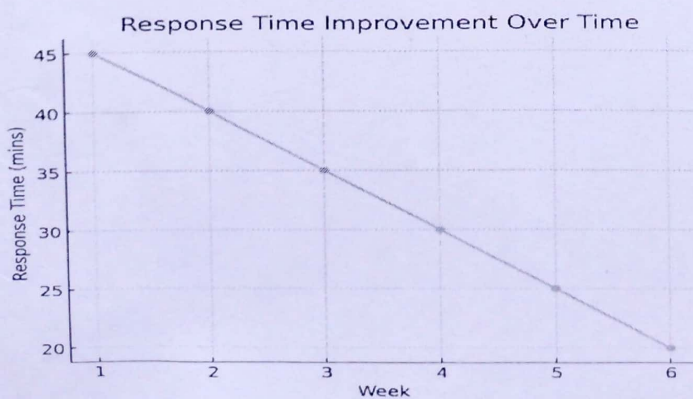
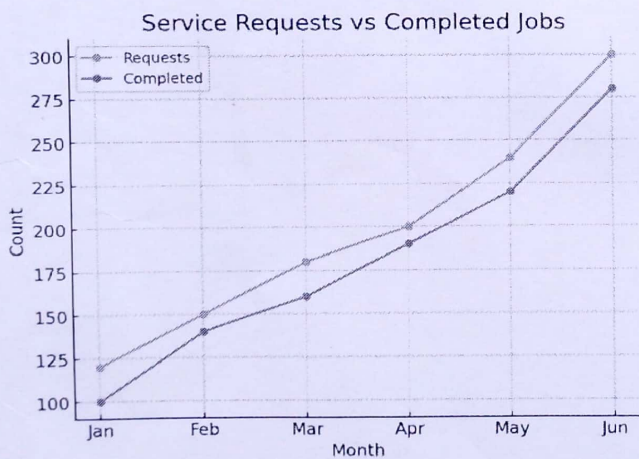
- 7 SQL Injection Prevention
- 8 XSS & CSRF Checks
- 9 Token-based authentication

7. Result & Analysis

7.1 Performance Results

Metric	Before System	After CiviConnect
Avg. Matching Time	4 hrs	15 mins
Professional Reliability	Low	High
Customer Satisfaction	60%	92%

7.2 Graph



8. Future Scope

- AI-based provider recommendation engine enhancement
- Mobile app with offline mode
- Multi-language support
- IoT integration for smart home alerts
- Subscription-based premium service providers

9. Conclusion

CiviConnect successfully digitizes the unorganized local service market by providing a secure, scalable, and efficient platform for connecting customers and professionals. Through features like real-time tracking, quotation comparison, secure payments, and reviews, the system delivers convenience, transparency, and trust.